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Valve seat with circumferential rib

Abstract

A valve seat includes an annular sealing body defining an inner seat surface and an outer seat surface, the inner seat surface defining an annular seating portion configured to seat with a valve member; and a circumferential sealing rib extending circumferentially about the annular sealing body, the circumferential sealing rib defining a gooseneck portion coupled to the annular sealing body and a bulbous portion coupled to the gooseneck portion opposite the annular sealing body.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to PCT Application No. CN2022/091171, filed May 6, 2022, which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) This disclosure relates to valves. More specifically, this disclosure relates to a high performance valve seat comprising a circumferential rib.

BACKGROUND

(3) Valves often have a valve member housed within a valve body, and a valve seat positioned between the valve member and the valve body to form a seal when the valve is placed in a closed configuration. Examples of common valve members include the ball of a ball valve, the disc of a butterfly valve, and the gate of a gate valve. The seat often is made of a resilient member that elastically deforms due to contact with the valve member to form the seal between the valve member and the seat. However, current valve seats are prone to failure, allowing fluid or gas to leak across the valve. Additionally, should the valve seat and/or the valve member become off-centered, abnormal wear on the valve seat can reduce the life cycle of the valve. Furthermore, opening and closing torques required to operate the valve often are undesirably high.

SUMMARY

(4) It is to be understood that this summary is not an extensive overview of the disclosure. This summary is exemplary and not restrictive, and it is intended to neither identify key or critical elements of the disclosure nor delineate the scope thereof. The sole purpose of this summary is to explain and exemplify certain concepts of the disclosure as an introduction to the following complete and extensive detailed description.

(5) Disclosed is a valve seat comprising an annular sealing body defining an inner seat surface and an outer seat surface, the inner seat surface defining an annular seating portion configured to seat with a valve member; and a circumferential sealing rib extending circumferentially about the annular sealing body, the circumferential sealing rib defining a gooseneck portion coupled to the annular sealing body and a bulbous portion coupled to the gooseneck portion opposite the annular sealing body.

(6) Also disclosed a valve comprising a valve body defining a main valve bore; a valve member disposed within the main valve bore and movable between an open orientation and a closed orientation, the valve member defining a seat engagement surface; a seat retainer mounted to the valve body; and a valve seat comprising an annular sealing body and a circumferential sealing rib, the circumferential sealing rib compressed between the valve body and the seat retainer, the annular sealing body defining an inner seat surface configured to seal with the seat engagement surface in the closed orientation.

(7) Additionally, disclosed is a valve seat comprising a sealing body defining a seating portion configured to seat with a valve member; and a sealing rib extending circumferentially about the annular sealing body, the sealing rib defining a bulbous portion.

(8) Various implementations described in the present disclosure may include additional systems, methods, features, and advantages, which may not necessarily be expressly disclosed herein but will be apparent to one of ordinary skill in the art upon examination of the following detailed description and accompanying drawings. It is intended that all such systems, methods, features, and advantages be included within the present disclosure and protected by the accompanying claims. The features and advantages of such implementations may be realized and obtained by means of the systems, methods, features particularly pointed out in the appended claims. These and other features will become more fully apparent from the following description and appended claims, or may be learned by the practice of such exemplary implementations as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features and components of the following figures are illustrated to emphasize the general

principles of the present disclosure. The drawings are not necessarily drawn to scale.

Corresponding features and components throughout the figures may be designated by matching reference characters for the sake of consistency and clarity.

(2) FIG. 1 is a perspective view of a valve comprising a valve member, a valve body, a seat retainer, and a high performance valve seat, in accordance with one aspect of the present disclosure.

(3) FIG. 2 is a cross-sectional view of the valve of FIG. 1, taken along line 2-2 shown in FIG. 1.

(4) FIG. 3A is a right-side view of the valve member of FIG. 1.

(5) FIG. 3B is a cross-sectional view showing a lower portion of the valve member of FIG. 3A, taken along line 2-2 shown in FIG. 1.

(6) FIG. 4 is a cross-sectional view showing a lower portion of the valve body of FIG. 1, taken along line 2-2 shown in FIG. 1.

(7) FIG. 5A is a perspective view of the seat retainer of FIG. 1.

(8) FIG. 5B is a cross-sectional view showing a lower portion of the seat retainer of FIG. 1, taken along line 2-2 shown in FIG. 1.

(9) FIG. 6A is a perspective view of the high performance valve seat of FIG. 1.

(10) FIG. 6B is a cross-sectional view showing a lower portion of the high performance valve seat of FIG. 1, taken along line 2-2 shown in FIG. 1.

(11) FIG. 7 is a front view of a spring of the valve of FIG. 1.

(12) FIG. 8 is a cross-sectional view of a lower portion of the valve of FIG. 1, taken along line 2-2 shown in FIG. 1.

(13) FIG. 9A is a cross-sectional view of a sealing rib of the high performance valve seat of FIG. 1, taken along line 2-2 shown in FIG. 1.

(14) FIG. 9B is a cross-sectional view of the sealing rib of the high performance valve seat in accordance with another aspect of the present disclosure.

(15) FIG. 10A is a cross-sectional view of a sealing body of the high performance valve seat, taken along line 2-2 shown in FIG. 1, wherein the spring of FIG. 7 is nested within a groove of the high performance valve seat.

(16) FIG. 10B is a cross-sectional view of the spring nested in the groove of the high performance valve seat, in accordance with another aspect of the present disclosure.

(17) FIG. 10C is a cross-sectional view of the sealing body of the high performance valve seat in accordance with another aspect of the present disclosure.

DETAILED DESCRIPTION

(18) The present disclosure can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and the previous and following description. However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this disclosure is not limited to the specific devices, systems, and/or methods disclosed unless otherwise specified, and, as such, can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

(19) The following description is provided as an enabling teaching of the present devices, systems, and/or methods in its best, currently known aspect. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made to the various aspects of the present devices, systems, and/or methods described herein, while still obtaining the beneficial results of the present disclosure. It will also be apparent that some of the desired benefits of the present disclosure can be obtained by selecting some of the features of the present disclosure without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present disclosure are possible and can even be desirable in certain circumstances and are a part of the present disclosure. Thus, the following description is provided as illustrative of the principles of the present disclosure and not in limitation thereof.

(20) As used throughout, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “an element” can include two or more such elements unless the context indicates otherwise.

(21) Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

(22) For purposes of the current disclosure, a material property or dimension measuring about X or substantially X on a particular measurement scale measures within a range between X plus an industry-standard upper tolerance for the specified measurement and X minus an industry-standard lower tolerance for the specified measurement. Because tolerances can vary between different materials, processes and between different models, the tolerance for a particular measurement of a particular component can fall within a range of tolerances.

(23) As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance can or cannot occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

(24) The word “or” as used herein means any one member of a particular list and also includes any combination of members of that list. Further, one should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain aspects include, while other aspects do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular aspects or that one or more particular aspects necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular aspect.

(25) Disclosed are components that can be used to perform the disclosed methods and systems. These and other components are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these components are disclosed, that while specific reference of each various individual and collective combinations and permutations of these may not be explicitly disclosed, each is specifically contemplated and described herein, for all methods and systems. This applies to all aspects of this application including, but not limited to, steps in disclosed methods. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific aspect or combination of aspects of the disclosed methods.

(26) Disclosed is a valve seat and associated methods, systems, devices, and various apparatus. The valve seat can comprise a sealing body and a sealing rib extending from the sealing body. It would be understood by one of skill in the art that the disclosed valve seat is described in but a few exemplary aspects among many. No particular terminology or description should be considered limiting on the disclosure or the scope of any claims issuing therefrom.

(27) FIG. 1 illustrates a valve **100** according to an example aspect of the present disclosure. In the present aspect, the valve **100** can be a butterfly valve **102**. However, in other aspects, the valve **100** can be any other suitable valve, including but not limited to, a ball valve, a gate valve, or the like. The butterfly valve **102** can define a first end **104** and a second end **106** opposite the first end **104**. According to example aspects, the butterfly valve **102** can comprise a valve body **110**, a valve member **140**, and a high performance valve seat **250** (shown in FIG. 2). The butterfly valve **102** can also comprise a seat retainer **160** configured to retain the high performance valve seat **250** on the valve body **110**. As shown, the butterfly valve **102** can further comprise a valve shaft **170** engaged with the valve member **140** and configured to move the valve member **140** between an open

orientation and a closed orientation (shown).

(28) The valve body **110** can extend from the first end **104** to the second end **106**, and can define an inner body surface **212** (shown in FIG. 2) and an outer body surface **114**. Each of the inner body surface **212** and the outer body surface **114** can be substantially cylindrical in the present aspect. The inner body surface **212** can define a main valve bore **116** extending through the valve body **110** from the first end **104** to the second end **106**. The valve body **110** can define a first bore opening **120** at the first end **104** of the valve **100** and a second bore opening **222** (shown in FIG. 2) at the second end **106** of the valve **100**. Each of the first bore opening **120** and the second bore opening **222** can allow access to the main valve bore **116**. The main valve bore **116** can define a main bore axis **118** extending centrally therethrough.

(29) In example aspects, the valve body **110** can further comprise one or more fastener lugs **124** extending radially outward from the outer body surface **114**. One or more mounting fastener holes **126** can be formed through each of the fastener lugs **124**, and a fastener, such as a bolt for example and without limitation, can extend through each corresponding mounting fastener hole **126** to mount the butterfly valve **102** to a surrounding structure, such as a pipeline. In other aspects, the valve body **110** can comprise a first annular flange at the first end **104** and a second annular flange at the second end **106**, and the mounting fastener holes **126** can be formed through the first and second annular flanges. According to example aspects, the valve body **110** can be formed by casting. For example, the valve body **110** can be formed from cast iron in example aspects. In other aspects, the valve body **110** can comprise any other suitable alloy or material, and can be formed by any suitable manufacturing process.

(30) According to example aspects, an annular seat retainer **160** can be disposed within the main valve bore **116** and can be coupled to the valve body **110**. The annular seat retainer can comprise a metal material in example aspects, such as, for example, steel. More specifically, in some aspects, the metal material can be WCB, CF8M, NAB, or the like. In other aspects, the annular seat retainer can comprise any other suitable metal or non-metal material. For example, in the present aspect, an inner body flange **214** (shown in FIG. 2) of the valve body **110** can extend radially inward from the inner body surface **212**, and the annular seat retainer **160** can be coupled to the inner body flange **214**. The annular seat retainer **160** can define one or more retainer fastener holes **162** formed therethrough. A fastener, such as a bolt for example and without limitation, can extend through each corresponding retainer fastener hole **162** and can engage the inner body flange **214** to couple the annular seat retainer **160** thereto. The annular seat retainer **160** can be disposed adjacent to the first end **104** of the valve **100**. However, in other aspects, the annular seat retainer **160** can be disposed adjacent to the second end **106** of the valve **100**, or at any suitable location within the main valve bore **116** between the first end **104** and the second end **106**. At least a portion of the high performance valve seat **250** can be sandwiched between the seat retainer **160** and the inner body flange **214**, as described in further detail below, to retain the high performance valve seat **250** on the valve body **110**.

(31) According to example aspects, the valve body **110** can further comprise a valve neck **128** extending from an upper one of the fastener lugs **124**. The butterfly valve **102** can comprise an upper gland flange **130** and a packing gland **234** (shown in FIG. 2) recessed within an upper body cavity **236** (shown in FIG. 2) of the valve neck **128**. As shown, the upper gland flange **130** can define a gland opening **132** configured to receive the valve shaft **170** therethrough. In example aspects, the upper gland flange **130** can be configured for mounting a valve actuator thereon, and the valve actuator can drive the movement of the valve member **140** between the open and closed orientations. The valve actuator can be, for example, a quarter-turn manual actuator, a gear box, or a motor, for example and without limitation. According to example aspects, mounting the valve actuator to the upper gland flange **130** can compress the packing gland **234** between the upper gland flange **130** and the valve body **110**, thereby energizing the packing gland **234**.

(32) The valve shaft **170** can extend vertically through the valve body **110** and the main valve bore

116, relative to the orientation shown. The valve shaft **170** can further extend through the valve member **140**. The valve member **140** can be a butterfly disc **142** in the present aspect. As shown, the valve member **140** can be positioned within the main valve bore **116**. The valve member **140** is shown in the closed orientation, wherein the valve member **140** can seal with the high performance valve seat **250**, as described in further detail below. In the closed orientation, the valve member **140** can block the main valve bore **116** and can prohibit fluid flow therethrough. The valve member **140** can be rotated about and between the open and closed orientations, respectively. In the open orientation, the valve member **140** can be disposed substantially sideways within the main valve bore **116**, allowing fluid to pass around the valve member **140** and through the main valve bore **116**. The valve shaft **170** can be fixedly coupled to the valve member **140**, such that rotating the valve shaft **170**, such as with the valve actuator, can rotate the valve member **140** about and between the open orientation and the closed orientation. In example aspects, the valve member **140** can comprise a substantially rigid material. For example, the valve member **140** can comprise a metal material in some aspects, such as, for example and without limitation, CF8M, NAB, MONEL, WCB, iron (including ductile iron), or the like. In other aspects, the valve member **140** can comprise any other suitable material.

(33) FIG. 2 illustrates a cross-sectional view of the butterfly valve **102** taken along line 2-2 in FIG. 1. As shown, the butterfly valve **102** can comprise the valve body **110**, the seat retainer **160**, the valve member **140**, the valve shaft **170**, and the high performance valve seat **250**. The upper gland flange **130** and the packing gland **234** can be disposed within the upper body cavity **236** of the valve neck **128**. An upper shaft channel **238** can extend through the valve body **110** from the upper body cavity **236** to the main valve bore **116**, and a lower shaft channel **240** can extend through the valve body **110** from a lower end **242** of the butterfly valve **102** to the main valve bore **116**. A lower channel cap **244** can be secured to the valve body **110** at the lower end **242**. The lower shaft channel **240** can be aligned with the upper shaft channel **238**, and the valve shaft **170** can extend through the upper body cavity **236** and the upper shaft channel **238**, across the main valve bore **116**, and into the lower shaft channel **240**. Moreover, the valve shaft **170** can be configured to rotate relative to the valve body **110** in order to rotate the valve member **140** about and between the open and closed orientations.

(34) In example aspects, the upper shaft channel **238** can define a widened portion **246** extending into the valve neck **128** from the upper body cavity **236**. A diameter of the widened portion **246** can be greater than a diameter of the upper shaft channel **238** between the widened portion **246** and the main valve bore **116**. One or more packings **260** can be received within the widened portion **246**. For example, in the present aspect, a plurality of cup and cone-style packings **262** can be stacked and mounted on the valve shaft **170** within the widened portion **246** and can be configured to seal the valve shaft **170** with the valve body **110**. The plurality of cup and cone-style packings **262** can allow for even compression against the valve shaft **170**. In example aspects, the cup and cone-style packings **262** can comprise PTFE polytetrafluoroethylene). However, in other aspects, the packing(s) **260** can be of any suitable type known in the art and/or can comprise any other suitable material, including but not limited to, graphite. According to example aspects, the packings **260** can be disposed between the packing gland **234** and an upper retaining ring **264**. The upper retaining ring **264** can be mounted on the valve shaft **170** and can abut a shoulder **248** of the widened portion **246**, as shown.

(35) The butterfly valve **102** can further comprise an upper bearing **268** and a lower bearing **270** configured to facilitate the rotation of the valve shaft **170** within the upper and lower shaft channels **238,240**. The upper bearing **268** can be mounted on the valve shaft **170** within the upper shaft channel **238**, such that the upper bearing **268** can be disposed between the upper retaining ring **264** and the main valve bore **116**. Similarly, the lower bearing **270** can be mounted on the valve shaft **170** within the lower shaft channel **240**, such that the lower bearing **270** can be disposed between a lower retaining ring **266** and the main valve bore **116**. According to example aspects, as shown, the

lower retaining ring **266** can be disposed within a lower channel groove **272** of the lower shaft channel **240** and a lower shaft groove **274** of the valve shaft **170**, adjacent to the lower channel cap **244**.

(36) According to example aspects, the valve member **140** (i.e., the butterfly disc **142**) can define a disc bore **276** through which the valve shaft **170** can extend, and the valve shaft **170** can be fixedly coupled to the butterfly disc **142**. As shown, a pair of tangential tapered holes **278** can be machined on a leading edge of the valve shaft **170** and on the disc bore **276**. A pin **280**, such as a taper pin **282** for example and without limitation, can be received in each of the tangential tapered holes **278**. To fixedly secure the valve shaft **170** to the butterfly disc **142**, each of the taper pins **282** can be welded to each of the valve shaft **170** and the butterfly disc **142** within the corresponding tangential tapered holes **278**. In other aspects, the butterfly valve **102** can comprise more or fewer of the tangential tapered holes **278** and the pins **280**. In other aspects, the valve shaft **170** can be fixedly coupled to the valve member **140** by any other suitable fastener or fastening technique known in the art.

(37) In example aspects, the annular seat retainer **160** can be substantially disposed within the main valve bore **116** at or near the first end **104** of the valve **100**. The inner body flange **214** of the valve body **110** can extend radially inward from the inner body surface **212**, and the annular seat retainer **160** can be coupled to the inner body flange **214**. At least a portion of the high performance valve seat **250** can be sandwiched between the seat retainer **160** and the inner body flange **214**, as shown, to retain the high performance valve seat **250** on the valve body **110**. For example, in the present aspect, the high performance valve seat **250** can comprise an annular sealing body **252** and a circumferential sealing rib **254** extending circumferentially about the annular sealing body **252**. The circumferential sealing rib **254** can be compressed between the seat retainer **160** and the inner body flange **214** of the valve body **110**. Furthermore, the valve member **140** (i.e., the butterfly disc **142**) can engage and seal with an inner seat surface **256** the annular sealing body **252** in the closed orientation, as described in additional detail below. According to example aspects, the high performance valve seat **250** can comprise relatively inflexible material, such as PTFE (i.e., polytetrafluoroethylene). In other aspects, the high performance valve seat **250** can comprise RTFE (reinforced PTFE), MRTFE, UHMWPE, or the like. In other aspects, the high performance valve seat **250** can comprise a combination of materials, such as PTFE & EPDM, PTFE & BUNA, PTFE & VITON, or the like, or any other suitable resilient, semi-resilient, or inflexible material or combination of materials known in the art. In some aspects, the valve seat **250** can comprise a combination of metal and non-metal materials, such as PTFE & 316SS, RPTFE & Inconel, or the like.

(38) FIG. 3A illustrates the valve member **140** (i.e., the butterfly disc **142**) in accordance with an example aspect of the present disclosure. As shown, the butterfly disc **142** can define a first disc end **302** and a second disc end **304** opposite the first disc end **302**. The butterfly disc **142** can generally define a seat engagement portion **306** formed at or near the first disc end **302** and a shaft engagement portion **308** extending from the seat engagement portion **306** and formed at or near the second disc end **304**. The seat engagement portion **306** of the butterfly disc **142** can define an annular seat engagement surface **310**, which can be configured to seal with the inner seat surface **256** of the high performance valve seat **250** (both shown in FIG. 2). In example aspects, the annular seat engagement surface **310** can define a substantially frustoconical shape, wherein a diameter of the annular seat engagement surface **310** can taper axially outwards, relative to the main bore axis **118** (shown in FIG. 1) in a direction towards the second disc end **304**. Furthermore, according to example aspects, the annular seat engagement surface **310** can be substantially convex. Specifically, the annular seat engagement surface **310** can be a spherical zone in the present aspect. In geometry, a spherical segment is the solid formed by cutting a sphere, or ball, with a pair of parallel planes. A spherical zone is the surface defined by the spherical segment.

(39) As shown in FIG. 3B, which illustrates a cross-sectional view of a lower portion of the

butterfly disc **142**, the shaft engagement portion **308** of the butterfly disc **142** can define the disc bore **276**, which can be configured to receive the valve shaft **170** (shown in FIG. **1**) therethrough. (40) FIG. **4** illustrates a cross-sectional view of a lower portion of the valve body **110**. As shown, the valve body **110** can define the inner body surface **212** and the outer body surface **114**. The inner body surface **212** can define the main valve bore **116**. The first bore opening **120** and the second bore opening **222** can allow access to the main valve bore **116**. The lower shaft channel **240** can extend through the valve body **110** from the inner body surface **212** to lower end **242** of the butterfly valve **102** (shown in FIG. **1**). Example aspects of the lower shaft channel **240** can define the lower channel groove **272**, within which the lower retaining ring **266** (shown in FIG. **2**) can be disposed. Moreover, the valve body **110** can define the inner body flange **214** extending radially inward from the inner body surface **212**. The inner body flange **214** can be disposed adjacent to the lower shaft channel **240**, between the lower shaft channel **240** and first bore opening **120**.

(41) The inner body flange **214** can define an annular front flange surface **402**, as shown. One or more flange fastener holes **404** can be formed through the front flange surface **402**. Each of the flange fastener holes **404** can be aligned with a corresponding one of the retainer fastener holes **162** (shown in FIG. **1**), and a corresponding one of the fasteners can extend through each corresponding retainer fastener hole **162** and flange fastener hole **404** to couple the annular seat retainer **160** (shown in FIG. **1**) to the inner body flange **214**. The inner body flange **214** can further define a top flange end **408**. The top flange end **408** can define a top flange surface **410**, and an annular flange rib **412** can extend radially inward from the top flange surface **410**. In example aspects, the top flange surface **410** can extend axially between the front flange surface **402** and the flange rib **412**, as shown. According to example aspects, the front flange surface **402** can further define an annular axially-indented body region **405** extending radially outward from the top flange surface **410**, and an annular body groove **406** connected to the axially-indented body region **405**, distal to the top flange surface **410**. The annular body groove **406** can define a substantially semi-circular cross-section in the present aspect.

(42) FIG. **5A** illustrates the annular seat retainer **160** and FIG. **5B** illustrates a cross-sectional view of a lower portion the annular seat retainer **160**, in accordance with an example aspect of the present disclosure. Referring to FIG. **5A**, the annular seat retainer **160** can define an inner retainer surface **502**, an outer retainer surface **504**, a front retainer surface **506** (shown in FIG. **5B**), and a rear retainer surface **508**. In some aspects, corners **510** are defined between intersecting ones of the inner retainer surface **502**, the outer retainer surface **504**, the front retainer surface **506**, and the rear retainer surface **508**. Each corner **510** can be a chamfered corner **510**. An annular upper notch **512** can be formed in the rear retainer surface **508** proximate to the inner retainer surface **502**. The upper notch **512** can at least partially define an annular inner retainer flange **514** at the inner retainer surface **502** and an annular outer retainer flange **516** opposite the annular inner retainer flange **514**. Each of the upper and outer retainer flange **516**s can extend axially away from the front retainer surface **506**. In example aspects, a substantially perpendicular portion **518** of the rear retainer surface **508** can extend substantially radially inward from the outer retainer surface **504** to the outer retainer flange **516**.

(43) Referring now to FIG. **5B**, in example aspects, the outer retainer flange **516** can define an angled upper flange surface **520**. For example, the angled upper flange surface **520** can be angled radially outward generally from the upper notch **512** to a distal flange end **522** of the outer retainer flange **516**. As shown, the distal flange end **522** of the outer retainer flange **516** can extend rearwardly beyond the substantially perpendicular portion **518**. Furthermore, according to example aspects, the angled upper flange surface **520** can be substantially convex. Specifically, the angled upper flange surface **520** can be a spherical zone in the present aspect. In example aspects, the substantially perpendicular portion **518** of the rear retainer surface **508** can define an annular axially-indented retainer region **524** extending radially outward from the outer retainer flange **516** and an annular retainer groove **526** connected to the axially-indented retainer region **524**, distal to

the outer retainer flange **516**. The annular retainer groove **526** can define a substantially semi-circular cross-section in the present aspect.

(44) FIG. **6A** illustrates a perspective view of the high performance valve seat **250** and FIG. **6B** illustrates a cross-sectional view of a lower portion of the high performance valve seat **250**, in accordance with an example aspect of the present disclosure. Referring to FIG. **6A**, the high performance valve seat **250** can comprise the annular sealing body **252** and the circumferential sealing rib **254** extending circumferentially about the annular sealing body **252**. The annular sealing body **252** can define a first seat end **602** and a second seat end **604**. Example aspects of the annular sealing body **252** can define an annular spring groove **606** defined between an inner sealing flange **608** and an outer sealing flange **610**. The spring groove **606** can be open towards the first seat end **602** and can be configured to receive an annular spring element **700** (shown in FIG. **7**) therein. According to example aspects, the annular spring groove **606** can provide increased open space for deformation of the high performance valve seat **250**. The increased space for deformation of the high performance valve seat **250** can reduce or eliminate the adverse effects of cold flow deformation (i.e., creep). With cold flow deformation, a valve seat can become permanently deformed due to the mechanical stresses imparted by a valve member. Allowing additional open space into which the high performance valve seat **250** can deform can reduce the mechanical stresses thereon, thereby reducing or eliminating cold flow deformation and prolonging the life cycle of the valve **100** (shown in FIG. **1**).

(45) Example aspects of the circumferential sealing rib **254** can comprise an annular gooseneck portion **612** and an annular bulbous portion **614**. A width of the bulbous portion **614** can be greater than a width of the gooseneck portion **612**. The gooseneck portion **612** can extend from substantially radially outward from an outer seat surface **620** (shown in FIG. **6B**) of the annular sealing body **252**, and the bulbous portion **614** can be coupled to the gooseneck portion **612** distal to the annular sealing body **252**. In the present aspect, the gooseneck portion **612** can extend from the annular sealing body **252** at or near the second seat end **604**.

(46) Referring to **6B**, the gooseneck portion **612** of the circumferential sealing rib **254** can define a first neck portion **616** extending substantially axially from the annular sealing body **252** at the second seat end **604** and towards the first seat end **602**. The gooseneck portion **612** can further define a second neck portion **618** extending substantially radially outward from the first neck portion **616**, distal to the annular sealing body **252**. The bulbous portion **614** of the circumferential sealing rib **254** can be coupled to the second neck portion **618** distal to the first neck portion **616**. In the present aspect, the bulbous portion **614** can define a substantially circular cross-section, as shown. In other aspects, the bulbous portion **614** can define a substantially rectangular cross-section (e.g., substantially square-shaped), as shown in FIG. **9B**. In other aspects, the bulbous portion **614** can define any other suitable cross-sectional shape. As described in further detail below, the gooseneck portion **612** and the bulbous portion **614** of the circumferential sealing rib **254** can be compressed between the seat retainer **160** (shown in FIG. **1**) and the inner body flange **214** (shown in FIG. **2**) of the valve body **110** (shown in FIG. **1**).

(47) The annular sealing body **252** of the circumferential sealing rib **254** can define the inner seat surface **256** and an outer seat surface **620**. Moreover, the annular sealing body **252** can comprise the inner sealing flange **608**, the outer sealing flange **610**, and the annular spring groove **606** defined therebetween. The annular spring groove **606** can define a substantially circular or rounded cross-sectional shape in the present aspect. In other aspects, the annular spring groove **606** can define a more rectangular cross-sectional shape, as shown in FIG. **10B**. In other aspects, the annular spring groove **606** can define any other suitable cross-sectional shape. Other aspects of the high performance valve seat **250** may not comprise the annular spring groove **606**, as shown in FIG. **10C**. Furthermore, in the present aspect, a distal inner flange end **622** of the inner sealing flange **608** can define a flange lip **624** arcing radially outwards, relative to the main valve bore **116**, towards the outer sealing flange **610**.

(48) According to example aspects, the inner seat surface **256** can at least partially define the inner sealing flange **608**. The inner seat surface **256** can further define an annular, angled seating portion **626** that can be angled radially inward generally in the direction from the second seat end **604** towards the first seat end **602**. In some aspects, the seating portion **626** can define a substantially frustoconical shape. In the present aspect, the seating portion **626** can optionally define a spherical zone. The spherical zone of the seat engagement portion **306** (shown in FIG. 3) of the butterfly disc **142** (shown in FIG. 1) can be configured to seat with the angled seating portion **626** of the high performance valve seat **250** in the closed orientation. Furthermore, the outer seat surface **620** can at least partially define the outer sealing flange **610**. In example aspects, the outer seat surface **620** can be angled radially inward from the gooseneck portion **612** of the circumferential sealing rib **254** to a distal outer flange end **628** of the outer sealing flange **610**. In some aspects, as shown, the outer sealing flange **610** as a whole can be angled radially inward in the direction from the first seat end **602** to the first seat end **602**. The outer sealing flange **610** can extend axially beyond the inner sealing flange **608** in the present aspect. The angled outer seat surface **620** of the high performance valve seat **250** can be configured to abut the angled upper flange surface **520** of the outer retainer flange **516** (both shown in FIG. 5B). Furthermore, according to example aspects, the outer seat surface **620** can be substantially convex. Specifically, the outer seat surface **620** can be shaped as a spherical zone, which can be complimentary to the spherical zone of the angled upper flange surface **520** (shown in FIG. 5B) of the outer retainer flange **516**.

(49) FIG. 7 illustrates the annular spring element **700** in accordance with an example aspect of the present disclosure. In the present aspect, the annular spring element **700** can be an annular helical spring **702** (i.e., a garter spring). In other aspects, the annular spring element **700** can be an annular cantilever spring **1000** (shown in FIG. 10B), which can define a substantially V-shaped cross-section or a substantially U-shaped cross-section, respectively. The annular helical spring **702** can be disposed within the annular spring groove **606** of the circumferential sealing rib **254** (both shown in FIG. 6B). The spring element **700** can comprise a resilient metal material in example aspects. In other aspects, the spring element **700** can comprise any other suitable material. According to example aspects, the annular spring element **700** can provide a continuous and consistent load force against the valve member **140** (shown in FIG. 1), which can ensure a stable and reliable seal between the valve member **140** and the high performance valve seat **250** (shown in FIG. 2).

(50) FIG. 8 illustrates a cross-sectional view of a lower portion of the valve **100** (i.e., the butterfly valve **102** in the present aspect). As shown, the butterfly valve **102** can comprise the valve body **110**, the seat retainer **160**, the valve member **140**, the valve shaft **170**, and the high performance valve seat **250**. The high performance valve seat **250** can be retained on the valve body **110** by sandwiching the circumferential sealing rib **254** between the seat retainer **160** and the inner body flange **214** of the valve body **110**. Specifically, the first neck portion **616** of the gooseneck portion **612** of the circumferential sealing rib **254** can be disposed and compressed between the top flange surface **410** of the inner body flange **214** and an underside **810** of the outer retainer flange **516** of the seat retainer **160**. The second neck portion **618** of the gooseneck portion **612** can be disposed and compressed between the axially-indented retainer region **524** of the seat retainer **160** and the axially-indented body region **405** of the inner body flange **214**. As shown, the axially-indented body and retainer regions **405,524** can together define a neck channel **812** therebetween for receiving the second neck portion **618**. Furthermore, the bulbous portion **614** of the circumferential sealing rib **254** can be disposed and compressed between the annular retainer groove **526** of the seat retainer **160** and the annular body groove **406** of the inner body flange **214**. Thus, the annular retainer groove **526** and the annular body groove **406** can together define a bulbous groove **814** therebetween for receiving the bulbous portion **614**. The bulbous groove **814** can define a substantially circular cross-sectional shape in the present aspect.

(51) According to example aspects, the valve member **140** can compress the bulbous portion **614** of

the circumferential sealing rib **254** into the bulbous groove **814**. The bulbous groove **814** can be configured to control, or limit, deformation of the high performance valve seat **250** within the bulbous groove **814** to create a compressed region **816** therein. The bulbous groove **814** can partially confine the compressed region **816** in both the radial and axial directions, thereby maintaining a high degree of compressive stress in the compressed region **816**. Specifically, the compressed region **816** can be maintained in a three-dimensional compressive state, which can provide a stable and reliable seal between the high performance valve seat **250** and the valve member **140**. Additionally, because the contact surface of the bulbous portion **614** with the valve body **110** and the seat retainer **160** is minimal, a lesser tightening force is required when securing the seat retainer **160** to the valve body **110** with the bulbous portion **614** sandwiched therebetween. (52) In example aspects, the second seat end **604** of the annular sealing body **252** can confront the annular flange rib **412** of the inner body flange **214**. The angled outer seat surface **620** of the annular sealing body **252** can extend along the angled upper flange surface **520** of the outer retainer flange **516**, and the distal outer flange end **628** of the outer sealing flange **610** can engage the upper notch **512** of the seat retainer **160**. According to example aspects, the spherical zone of the outer seat surface **620** can be mated with the spherical zone of the upper flange surface **520**. Contact between the convex shape of the outer seat surface **620** and the complimentary concave shape of the upper flange surface **520** can reduce the rotational torque from friction required to rotate the valve member **140** against the high performance valve seat **250**. Additionally, the concave shape of the upper flange surface **520** can prevent the high performance valve seat **250** from shifting when the valve **100** is pressurized because the concave upper flange surface **520** can receive and center the convex shape of the outer seat surface **620**.

(53) The annular helical spring **702** can be disposed in the spring groove **606** between the inner sealing flange **608** and the outer sealing flange **610**. As shown, the seat engagement portion **306** of the butterfly disc **142** can engage and seal with the inner seat surface **256** of the annular sealing body **252** to prohibit fluid flow through the main valve bore **116** in the closed orientation of the butterfly disc **142**. Specifically, the annular seat engagement surface **310** of the seat engagement portion **306** of the butterfly disc **142** can engage and seal with the angled seating portion **626** of the inner seat surface **256** of the high performance valve seat **250**. The annular sealing body **252** can be compressed between the butterfly disc **142** and the seat retainer **160** in the closed orientation, and the annular helical spring **702** can provide continuous and consistent resistance to the compression, thereby improving the seal between the high performance valve seat **250** and the butterfly disc **142** and between the high performance valve seat **250** and the seat retainer **160**.

(54) According to example aspects, the spherical zone of the seat engagement surface **310** of the valve member **140** (i.e., the butterfly disc **142**) can be mated with the spherical zone of the angled seating portion **626** of the inner seat surface **256** of the high performance valve seat **250**. Contact between the convex shape of the seat engagement surface **310** and the concave shape of the seating portion **626** of the inner seat surface **256** can reduce the rotational torque from friction required to rotate the valve member **140** against the high performance valve seat **250**. Additionally, the concave shape of the angled seating portion **626** can prevent the high performance valve seat **250** and the valve member **140** from shifting when the valve **100** is pressurized because the concave portion can receive and center the convex shape of the seat engagement surface **310**.

(55) FIGS. **9A** and **9B** illustrate different cross-sectional shapes of the bulbous portion **614** of the circumferential sealing rib **254** of the high performance valve seat **250**. In one aspect, as shown in FIG. **9A**, the bulbous portion **614** can define a substantially circular cross-sectional shape. In another aspect, as shown in FIG. **9B**, the bulbous portion **614** can define a substantially rectangular cross-sectional shape. Specifically, the cross-sectional shape of the bulbous portion **614** can be substantially square-shaped in the present aspect. In some aspects, a lower surface **900** of the bulbous portion **614** can define a first angled surface portion **902** and a second angled surface portion **904**. Each of the first and second angled surface portions **902,904** can be angled slightly

outward from the bulbous portion **614** and can meet at an apex **906**, as shown. In other aspects, the bulbous portion **614** of the circumferential sealing rib **254** can define any other suitable cross-sectional shape.

(56) FIGS. **10A-10C** illustrate different cross-sectional shapes of the annular sealing body **252** of the circumferential sealing rib **254** of the high performance valve seat **250**. In one aspect, as shown in FIG. **10A**, the annular sealing body **252** can define the inner seat surface **256** and the outer seat surface **620**. The annular sealing body **252** can comprise the inner sealing flange **608**, the outer sealing flange **610**, and the annular spring groove **606** defined therebetween. The annular spring groove **606** can be configured to receive the annular spring element **700** therein. The annular spring groove **606** can define a substantially circular cross-sectional shape. The annular spring element **700** can be the annular helical spring **702**, which can be disposed within the annular spring groove **606** of the circumferential sealing rib **254**. In some aspects, as shown, the distal inner flange end **622** of the inner sealing flange **608** can define the flange lip **624** arcing radially outwards, relative to the main valve bore **116** (shown in FIG. **1**) towards the outer sealing flange **610**.

(57) In another aspect, as shown in FIG. **10B**, the annular spring groove **606** can define a less circular and more rectangular cross-sectional shape. In the present aspect, inner corners **1002** of the annular spring groove **606** can be rounded, such that the annular spring groove **606** can define a substantially U-shaped cross-sectional shape. The annular spring element **700** disposed within the annular spring groove **606** can be an annular cantilever spring **1000** defining a substantially U-shaped cross-section. In example aspects, the annular cantilever spring **1000** can define an annular inner free end **1004** configured to engage an annular inner shoulder **1008** of the high performance valve seat **250** and an annular outer free end **1006** configured to engage an annular outer shoulder **1010** of the high performance valve seat **250** to retain the cantilever spring **1000** within the spring groove **606**. Furthermore, in the present aspect, the inner sealing flange **608** can be substantially linear and does not define the flange lip **624** arcing towards the outer sealing flange **610**.

(58) In other aspects, the annular spring groove **606** can define a substantially V-shaped cross-sectional shape and/or the spring element **700** disposed within the annular spring groove **606** can be an annular cantilever spring **1000** defining a substantially V-shaped cross-sectional shape. In other aspects, the annular spring groove **606** and the spring element **700** can define any other suitable cross-sectional shape.

(59) In other aspects, such as the aspect depicted in FIG. **10C**, the annular sealing body **252** may not define the annular spring groove **606**. The annular sealing body **252** of the present aspect further does not define the inner sealing flange **608** and the outer sealing flange **610** (both shown in FIG. **6A**). Rather, the annular sealing body **252** can define an engagement lip **1010** at the first seat end **602**. In the present aspect, the engagement lip **1010** can be partially defined by the outer seat surface **620**. Similar to the distal outer flange end **628** (shown in FIG. **6B**) of the outer sealing flange **610**, the engagement lip **1010** can be configured to engage the annular upper notch **512** (shown in FIG. **5A**) of the seat retainer **160** (shown in FIG. **1**). The annular sealing body **252** can further define a substantially planar first end surface **1012** formed at the first seat end **602** and extending between the engagement lip **1010** and the inner seat surface **256**.

(60) According to example aspects, a method of installing the high performance valve seat **250** with the valve **100** can comprise disposing the circumferential sealing rib **254** of the high performance valve seat **250** between the valve body **110** and the seat retainer **160**. The method can further comprise fastening the seat retainer **160** to the valve body **110** to compress the circumferential sealing rib **254** therebetween. In some aspects, the circumferential sealing rib **254** can define the bulbous portion **614**, the valve body **110** and the seat retainer **160** can define the bulbous groove **814** therebetween, and disposing the circumferential sealing rib **254** between the valve body **110** and the seat retainer **160** can comprise disposing the bulbous portion **614** within the bulbous groove **814**. Furthermore, in some aspects, the circumferential sealing rib **254** can further define the gooseneck portion **612**, the valve body **110** and the seat retainer **160** can further define the neck

channel **812** therebetween, and disposing the circumferential sealing rib **254** between the valve body **110** and the seat retainer **160** can further comprise disposing the second neck portion **618** of the gooseneck portion **612** within the neck channel **812**. In some aspects, fastening the seat retainer **160** to the valve body **110** to compress the circumferential sealing rib **254** therebetween can comprise receiving a fastener through each corresponding pair of retainer fastener holes **162** and flange fastener holes **404** and tightening the fasteners to draw the seat retainer **160** towards the valve body **110**.

(61) Additionally, a method of operating the valve **100** can comprise moving the valve member **140** from the open orientation to the closed orientation and sealing the valve member **140** with the high performance valve seat **250** in the closed orientation to prohibit fluid flow through the valve **100**. In example aspects, sealing the valve member **140** with the high performance valve seat **250** can comprise sealing the inner seat surface **256** of the high performance valve seat **250** with the seat engagement surface **310** of the valve member **140**. The method can further comprise moving the valve member **140** from the closed orientation to the open orientation and unsealing the valve member **140** from the high performance valve seat **250** to permit fluid flow through the valve **100**.

(62) In one exemplary aspect, a valve seat can comprise an annular sealing body that can define an inner seat surface and an outer seat surface, and the inner seat surface can define an annular seating portion configured to seat with a valve member. The valve seat can further comprise a circumferential sealing rib that can extend circumferentially about the annular sealing body, and the circumferential sealing rib can define a gooseneck portion coupled to the annular sealing body and a bulbous portion coupled to the gooseneck portion opposite the annular sealing body.

(63) In a further exemplary aspect, the valve seat can define an axis extending centrally therethrough, and the circumferential sealing rib can extend substantially radially outward from the annular sealing body. In a further exemplary aspect, the gooseneck portion of the circumferential sealing rib can define a first neck portion and a second neck portion, the first neck portion can extend from the annular sealing body in a substantially axial direction, and the second neck portion can extend substantially radially outward from the first neck portion, about perpendicular to the first neck portion. In a further exemplary aspect, a width of the bulbous portion can be greater than a width of the gooseneck portion. In a further exemplary aspect, the bulbous portion can define a substantially circular cross-sectional shape. In a further exemplary aspect, the bulbous portion can define a substantially rectangular cross-sectional shape. In a further exemplary aspect, the outer seat surface of the annular sealing body can define a spherical zone. In a further exemplary aspect, the annular seating portion can define a spherical zone. In a further exemplary aspect, the annular sealing body can define an inner sealing flange and an outer sealing flange, an annular spring groove can be defined between the inner sealing flange and the outer sealing flange, and the annular spring groove can be configured to receive a spring therein. In a further exemplary aspect, the outer seat surface can at least partially define the outer sealing flange, and the inner seat surface can at least partially define the inner sealing flange. In a further exemplary aspect, the outer sealing flange can extend axially beyond the inner sealing flange. In a further exemplary aspect, the spring can be one of an annular helical spring and an annular cantilever spring.

(64) In another exemplary aspect, a valve can comprise a valve body that can define a main valve bore, a valve member that can be disposed within the main valve bore and can be movable between an open orientation and a closed orientation, a seat retainer that can be mounted to the valve body, and a valve seat that can comprise an annular sealing body and a circumferential sealing rib. At least a portion of the circumferential sealing rib can be compressed between the valve body and the seat retainer. The valve member can define a seat engagement surface, and the annular sealing body can define an inner seat surface configured to seal with the seat engagement surface in the closed orientation.

(65) In a further exemplary aspect, the valve body can define an annular body groove, the seat retainer can define an annular retainer groove, the annular body groove and the annular retainer

groove can together define a bulbous groove, and the circumferential sealing rib can define a bulbous portion compressed within the bulbous groove. In a further exemplary aspect, the circumferential sealing rib can further define a gooseneck portion coupled to the annular sealing body, the bulbous portion can be coupled to the gooseneck portion opposite the annular sealing body, the valve body can define an annular axially-indented body region connected to the annular body groove, the seat retainer can define an annular axially-indented retainer region connected to the annular retainer groove, and the annular axially-indented body region and the annular axially-indented retainer region can together define a neck channel within which at least a portion of the gooseneck portion can be compressed. In a further exemplary aspect, the gooseneck portion of the circumferential sealing rib can define a first neck portion extending from the annular sealing body in a substantially axial direction; the gooseneck portion can further define a second neck portion extending substantially radially outward from the first neck portion and about perpendicular to the first neck portion, and the second neck portion can be compressed within the neck channel. In a further exemplary aspect, the bulbous portion can define a substantially circular cross-sectional shape. In a further exemplary aspect, the bulbous portion can define a substantially rectangular cross-sectional shape. In a further exemplary aspect, the annular sealing body can further define an outer seat surface, and the outer seat surface can define a convex spherical zone. The seat retainer can define a retainer flange, and the retainer flange can define an upper flange surface. The upper flange surface can define a concave spherical zone complimentary to and receiving the convex spherical zone of the outer seat surface. In a further exemplary aspect, the inner seat surface can define an annular seating portion configured to seal with the seat engagement surface of the valve member in the closed orientation; the annular seating portion can define a concave spherical zone, and the seat engagement surface can define a convex spherical zone complimentary to the concave spherical zone of the annular seating portion. In a further exemplary aspect, the annular sealing body can define an inner sealing flange and an outer sealing flange, an annular spring groove can be defined between the inner sealing flange and the outer sealing flange, and the valve can further comprise an annular spring received within the annular spring groove. In a further exemplary aspect, the outer seat surface can at least partially define the outer sealing flange, a distal outer flange end of the outer sealing flange can engage an annular notch of the seat retainer, and the annular notch can be at least partially defined by the upper flange surface of the retainer flange.

(66) In another exemplary aspect, a valve seat can comprise a sealing body defining a seating portion configured to seat with a valve member. The valve seat can further comprise a sealing rib extending circumferentially about the annular sealing body. The sealing rib can define a bulbous portion.

(67) One should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular embodiments or that one or more particular embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

(68) It should be emphasized that the above-described embodiments are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the present disclosure. Any process descriptions or blocks in flow diagrams should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process, and alternate implementations are included in which functions may not be included or executed at all, may be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those reasonably

skilled in the art of the present disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the present disclosure. Further, the scope of the present disclosure is intended to cover any and all combinations and sub-combinations of all elements, features, and aspects discussed above. All such modifications and variations are intended to be included herein within the scope of the present disclosure, and all possible claims to individual aspects or combinations of elements or steps are intended to be supported by the present disclosure.

Claims

1. A valve seat comprising: an annular sealing body defining an inner sealing flange defining an inner seat surface, an outer sealing flange defining an outer seat surface, and an annular spring groove defined between the inner sealing flange and the outer sealing flange, the inner seat surface defining an annular seating portion configured to seat with a valve member; and a circumferential sealing rib extending circumferentially about the annular sealing body, the circumferential sealing rib defining a gooseneck portion coupled to the annular sealing body and a bulbous portion coupled to the gooseneck portion opposite the annular sealing body, wherein the outer seat surface of the annular sealing body defines a spherical zone.
2. The valve seat of claim 1, wherein: the valve seat defines an axis extending centrally therethrough; and the circumferential sealing rib extends substantially radially outward from the annular sealing body.
3. The valve seat of claim 2, wherein: the gooseneck portion of the circumferential sealing rib defines a first neck portion and a second neck portion; the first neck portion extends from the annular sealing body in a substantially axial direction; and the second neck portion extends substantially radially outward from the first neck portion, about perpendicular to the first neck portion.
4. The valve seat of claim 2, wherein a width of the bulbous portion is greater than a width of the gooseneck portion.
5. The valve seat of claim 4, wherein the bulbous portion defines a substantially circular cross-sectional shape.
6. The valve seat of claim 4, wherein the bulbous portion defines a substantially rectangular cross-sectional shape.
7. The valve seat of claim 1, wherein the annular seating portion defines a spherical zone.
8. The valve seat of claim 7, wherein the annular spring groove is configured to receive a spring therein.
9. The valve seat of claim 8, wherein the outer sealing flange extends axially beyond the inner sealing flange.
10. The valve seat of claim 8, wherein the spring is one of an annular helical spring and an annular cantilever spring.
11. A valve seat comprising: an annular sealing body defining an inner seat surface and an outer seat surface, the inner seat surface defining an annular seating portion configured to seat with a valve member; and a circumferential sealing rib extending circumferentially about the annular sealing body, the circumferential sealing rib defining a gooseneck portion coupled to the annular sealing body and a distal end portion coupled to the gooseneck portion opposite the annular sealing body; wherein the outer seat surface of the annular sealing body defines a spherical zone; wherein the annular seating portion defines a spherical zone; and wherein the annular sealing body defines an inner sealing flange and an outer sealing flange, an annular spring groove defined between the inner sealing flange and the outer sealing flange, the annular spring groove configured to receive a spring therein.
12. The valve seat of claim 11, wherein: the valve seat defines an axis extending centrally

therethrough; and the circumferential sealing rib extends substantially radially outward from the annular sealing body.

13. The valve seat of claim 12, wherein: the gooseneck portion of the circumferential sealing rib defines a first neck portion and a second neck portion; the first neck portion extends from the annular sealing body in a substantially axial direction; and the second neck portion extends substantially radially outward from the first neck portion, about perpendicular to the first neck portion.

14. The valve seat of claim 12, wherein a width of the distal end portion is greater than a width of the gooseneck portion.

15. The valve seat of claim 14, wherein the distal end portion defines a substantially circular cross-sectional shape.

16. The valve seat of claim 14, wherein the distal end portion defines a substantially rectangular cross-sectional shape.

17. The valve seat of claim 11, wherein the outer seat surface at least partially defines the outer sealing flange, and wherein the inner seat surface at least partially defines the inner sealing flange.

18. The valve seat of claim 11, wherein the outer sealing flange extends axially beyond the inner sealing flange.

19. The valve seat of claim 11, wherein the spring is one of an annular helical spring and an annular cantilever spring.
